Answers and talk notes

Understand Equations and Their Solutions

What your student is learning

Tonight we turn a word problem into an equation — naming the unknown, writing the sentence in symbols, and testing a guess.

Answers

Ask your student to explain before you look.

1. **B.** a number increased by 8. The phrase a number increased by 8 means n + 8.
2. **A.** n − 8. n − 8 is correct because the number comes first and 8 is taken away.
3. **C.** n ÷ 4 = 6. n ÷ 4 = 6 puts the unknown number first and divides it by 4.
4. **D.** 2n = 18. 2n = 18 means 2 times the number is 18.
5. **A.** 20 − m = 13. 20 − m = 13 shows starting with 20 and taking away m.
6. **C.** Yes, because 9 − 4 = 5. Yes, because 9 − 4 = 5, so both sides match.

Talk about it

What does your letter stand for? Which number comes first in the symbols, and why?

Home mission

Count out 12 spoons or coins. Hide some under a cup and leave 7 in view. Can your student write 12 − c = 7? Then lift the cup to test a guess.

When they are stuck, ask

- What are we trying to find? Give it a letter.
- Can you say the relationship in words before using symbols?
- Try a number. Was it too big or too small?

Try not to say

- Write the numbers in the order they appear in the sentence.
- Guessing is not real math, just solve it.
- "Decreased by 8" means 8 minus the number.